AI-POWERED E-WASTE SORTING AND BIOLEACHING SIMULATOR

A Waste-to-Wealth Project

Subject	Environmental Biotechnology
Project focus	Artificial Intelligence, Electronics, and Sustainable Metal Recovery
Implementation	Free software-based simulation; no laboratory work required

Project concept

This project proposes an AI-assisted workflow that classifies electronic waste from images and simulates the potential recovery of valuable metals—especially copper—from printed circuit board (PCB) waste using bioleaching as the environmental biotechnology component.

The prototype is designed to be achievable with free tools and hypothetical simulation inputs. It does not require purchasing electronic waste, chemicals, sensors, or microorganisms.

Prepared as a project proposal / prototype plan

1. Problem Statement

Electronic waste contains valuable materials such as copper, but ineffective sorting and improper disposal can cause resource loss and environmental pollution. Manual identification of e-waste can be slow, and metal recovery from circuit boards requires appropriate processing methods. This project proposes an AI-assisted system that identifies e-waste categories from images and simulates potential metal recovery through bioleaching. The aim is to demonstrate how image classification and environmentally conscious resource recovery could help convert e-waste into a useful resource.

2. Aim and Objectives

Aim: To design a free, software-based prototype that combines AI-based e-waste classification with a simulation of copper recovery through bioleaching.

Objectives:

- Classify e-waste images into broad categories such as circuit boards, batteries, cables, and plastic casings.
- Identify circuit-board waste as a candidate stream for metal recovery.
- Use adjustable assumptions to estimate the mass of copper potentially recovered.
- Show results in a simple dashboard or graph.
- Explain the role and limitations of microbial bioleaching in resource recovery.

3. Integrated Workflow

Step 1 — Image input

Upload or select an image of e-waste, such as a circuit board, battery, cable, or plastic casing.

Step 2 — AI image classification

A classifier predicts the broad category of the item. The output should be treated as a prediction, not a guaranteed identification.

Step 3 — Recovery route selection

If the item is classified as circuit-board waste, the prototype opens the copper-recovery simulation. Other categories can be flagged for their appropriate recycling stream.

Step 4 — Bioleaching simulation

The user enters the waste mass, assumed copper content, and assumed recovery efficiency. The simulator estimates potential recovered copper.

Step 5 — Results dashboard

Display the predicted waste category, input assumptions, estimated copper recovery, and a chart comparing different scenarios.

4. Example Simulation and Calculation

Assume the prototype receives a batch of circuit-board waste. The values below are illustrative assumptions for demonstrating the calculation; they are not experimental measurements.

Parameter	Example value
Circuit-board waste mass	1,000 g (1 kg)

Assumed copper content	15% by mass
Assumed bioleaching recovery efficiency	60%

Formula

Estimated copper recovered = waste mass × copper fraction × recovery efficiency

= 1,000 g × 0.15 × 0.60 = 90 g

The result means that, under these assumed conditions, the model estimates 90 g of copper could be recovered from 1 kg of circuit-board waste. Actual recovery depends on the waste composition, microorganisms or leaching system, operating conditions, and process performance.

5. Tools and Resources

Tool / resource	Purpose
Google Colab	Run Python code and the calculation/simulation at no cost within available usage limits.
Python (pandas, matplotlib)	Organize inputs and plot recovery scenarios.
Public e-waste image dataset or selected sample images	Provide images for classification demonstrations.
Pre-trained image classifier or a small trained model	Predict a broad e-waste category; use only if suitable and document limitations.
Optional: Streamlit	Create an interactive interface if time and environment permit.

For the fastest one-day prototype, the core deliverable can be a notebook with a simple category-selection interface, recovery calculator, and graph. Image classification can be added using an available model or a small dataset if time permits.

6. Suggested Prototype Interface

The prototype can present the following outputs on one screen:

- Image preview and predicted category
- Category confidence, if the chosen model provides a valid confidence score
- Waste mass, assumed copper content, and assumed recovery efficiency
- Estimated copper recovered in grams
- A graph comparing recovery efficiency values such as 20%, 40%, 60%, and 80%

7. Expected Outcomes

The completed prototype is expected to demonstrate how AI-assisted classification can support e-waste sorting and how a simple model can estimate possible resource recovery. It should also help explain the potential role of bioleaching as an alternative or complementary metal-recovery approach. The outputs are educational estimates, not proof that copper has actually been recovered.

8. Environmental Significance

Better identification and segregation of e-waste can help direct materials toward suitable recycling or treatment pathways. Bioleaching uses biological activity to mobilize metals from certain waste streams under controlled conditions. The project connects these ideas through a digital demonstration of waste identification, recovery estimation, and resource conservation. The simulation does not establish that bioleaching is always safer, cheaper, or more efficient than other methods; those claims would require evidence and comparison.

9. Limitations and Precautions

- Image classification can fail when images are unclear, unusual, or outside the model's training categories.
- Copper content and recovery efficiency are assumed unless reliable measured data are supplied.
- The simulation estimates potential recovery; it does not conduct actual bioleaching.
- Real e-waste processing can involve hazardous materials. Do not dismantle batteries or circuit boards or carry out chemical extraction without appropriate supervision, facilities, and safety procedures.

10. Conclusion

The AI-Powered E-Waste Sorting and Bioleaching Simulator combines image classification with a simple environmental biotechnology model. By classifying e-waste and estimating potential copper recovery from circuit boards, it demonstrates a possible digital pathway from waste identification to resource recovery. The project is suitable as a low-cost proof of concept, with clear opportunities for future improvement using validated datasets and experimentally measured recovery data.

Suggested references for background reading

1. Applied Sciences (2025), research on an AI-driven RGB-based approach for copper recovery via printed circuit board bioleaching: https://www.mdpi.com/2076-3417/15/1/129

2. Sustainability (2024), research on bioleaching of printed circuit board waste: https://www.mdpi.com/2071-1050/16/22/9837

3. PubMed Central (2022), mathematical modeling of metal recovery from e-waste via leaching: https://pmc.ncbi.nlm.nih.gov/articles/PMC8917181/

Note: Check the final bibliographic details and relevance of each source before submitting. Clearly label all simulated values and model outputs in your report.